If V is an (A, B)-bimodule and W is a (B, C)-bimodule, then these two structures on $V \otimes_B W$ can be combined into one (A, C)-bimodule structure on $V \otimes_B W$.

(a) Let A, B, C, D be four algebras. Let V be an (A, B)-bimodule, W a (B, C)-bimodule, and X a (C, D)-bimodule. Prove that $(V \otimes_B W) \otimes_C X \cong V \otimes_B (W \otimes_C X)$ as (A, D)-bimodules. The isomorphism (from left to right) is given by the formula

$$(v \otimes_B w) \otimes_C x \mapsto v \otimes_B (w \otimes_C x)$$

for all $v \in V$, $w \in W$, and $x \in X$.

(b) If A, B, C are three algebras and if V is an (A, B)-bimodule and W an (A, C)-bimodule, then the vector space $\operatorname{Hom}_A(V, W)$ (the space of all left A-linear homomorphisms from V to W) canonically becomes a (B, C)-bimodule by setting $(bf)(v) = f(vb)$ for all $b \in B$, $f \in \operatorname{Hom}_A(V, W)$, and $v \in V$ and setting $(fc)(v) = f(v)c$ for all $c \in C$, $f \in \operatorname{Hom}_A(V, W)$ and $v \in V$.

Let A, B, C, D be four algebras. Let V be a (B, A)-bimodule, W a (C, B)-bimodule, and X a (C, D)-bimodule. Prove that

$$\operatorname{Hom}_B(V, \operatorname{Hom}_C(W, X)) \cong \operatorname{Hom}_C(W \otimes_B V, X)$$

as (A, D)-bimodules. The isomorphism (from left to right) is given by

$$f \mapsto (w \otimes_B v \mapsto f(v)w)$$

for all $v \in V$, $w \in W$ and $f \in \operatorname{Hom}_B(V, \operatorname{Hom}_C(W, X))$.

Exercise 2.11.7. Show that if M and N are modules over a commutative ring A, then $M \otimes_A N$ has a natural structure of an A-module.

2.12. The tensor algebra

The notion of tensor product allows us to give more conceptual (i.e., coordinate-free) definitions of the free algebra, polynomial algebra, exterior algebra, and universal enveloping algebra of a Lie algebra.

Namely, given a vector space V, define its **tensor algebra** TV over a field k to be $TV = \bigoplus_{n \geq 0} V^{\otimes n}$, with multiplication defined by $a \cdot b := a \otimes b$, $a \in V^{\otimes n}$, $b \in V^{\otimes m}$. Observe that a choice of a basis